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Three Notions of Linearization for a Discrete Ising System: Operator, Statistical, and Projection-Based Approximations

Summary:

This project investigates different methods of approximating nonlinear systems that contain stochastic components, specifically focusing on how traditional local linearization via a Jacobian fails to capture high-dimensional behaviors. The paper argues that for such systems, linearization is an ambiguous modeling choice that leads to multiple non-equivalent representations. The project uses the Ising model as a test case to see how well the different methods can approximate the system's behavior.

Introduction: (5/5)

- This paper clearly describes the motivation of the in the first section.
- Thorough background and related work.
- Aim of the project is well stated.

Professionalism: (4.5/5)

- Paper is clearly written in a standard template format.
- Objectives are clear.
- Readability issues (-0.5):
  - In paragraph 4, the sentence, "Although the Ising model was first proposed to model the spins of atoms in a material, the properties of the model, including phase transitions and a critical [3], it and essentially isomorphic models are used to model percolation processes [4], phase transitions [5], and forest fires [6], to name only a few. All of these modifications and settings display the same scale-free behavior, phase transitions, and criticality." is unclear.
  - Paragraph 3 and 4 have extremely long sentences.
  - Content from paragraph 4 can be incorporated into near the beginning of paragraph 3 and then paragraph 3 can be split into two separate paragraphs.
- Additionally, some are the References are youtube videos. Didn't deduct points for this.